REFINING A ONE STEP DEVISSAGE ALONG AN ELEMENTARY ETALE NEIGHBOURHOOD OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed. The sentence 'Of course in general this is not going to be true' and the surrounding argument at proof lines 1116-1134 are the author's own account of the only obstruction.

Editorial note. The two leading context spans are the author's own definitions of a one step devissage over s and at x, printed first because the counted statement opens with a verbatim reference to the second of them; they define the labels definition-one-step-devissage and definition-one-step-devissage-at-x, which therefore do not appear among the external references.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros. The native class line, the xr-hyper externaldocument block and the reference/slogan/history display environments are omitted because the assembler supplies a plain amsart wrapper and the excerpt uses none of those environments; no macro was invented or redefined.

Editorial note. Locators of the cross-chapter references point either at the \label line of the referenced result inside the pinned chapter file (same-chapter reference 057R) or at a verbatim excerpt of the project's own tag list, sources/stacks-flat/referenced-tags.txt, each recorded with its stable Stacks tag.

Allowed standard prerequisites (editorial).

- Elementary etale neighbourhoods of pointed schemes. The statement is about an elementary etale neighbourhood (Y', y') -> (Y, y) -> (S, s) and produces an elementary etale neighbourhood (X", x") -> (X, x); the construction of the new diagram is the retained obligation.
- Invariance of the dimension of a support under etale localization. Used to keep dim_{x"} Supp(F"_{s"}) equal to n; the surrounding verifications of the definition are retained.
- Existence of an elementary etale neighbourhood making all fibres geometrically irreducible. Supplies the final (S", s") -> (S', s') and affine Y" once the proof itself has produced the smooth geometrically connected component T; the reduction to that component and the residue-field argument are retained.

Source reference. Original lemma 02KG: sources/stacks-flat/referenced-tags.txt, source lines 5–5. Cross-chapter pointer used at proof lines 1084-1087 to identify the pushforward of G" with F" over affine schemes; it records an identification the counted proof itself sets up by construction.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04N4: sources/stacks-flat/referenced-tags.txt, source lines 6–6. Cross-chapter standard prerequisite (dimension at a point is etale local) invoked at proof lines 1099-1107 to keep the fibre support dimension equal to n; the residue-field and etale comparison argument around it is retained.

Source reference. Original lemma 057R: sources/stacks-flat/pinned-original-source.tex, source lines 84–84. Same-chapter pointer to the etale lifting lemma invoked at proof lines 1072-1074 to build X" with Z" = X" x_{X_{S'}} Z_{S'}; the choice of Y", S" and the verification of all the devissage conditions remain in the counted proof.

Source reference. Original lemma 057J: sources/stacks-flat/referenced-tags.txt, source lines 7–7. Cross-chapter input invoked at proof lines 1127-1132 for the final choice of (S", s") and Y"; the proof's own reduction to the smooth component T and its residue-field argument are retained and are what make the pointer applicable.

Source reference. Original lemma 04KV: sources/stacks-flat/referenced-tags.txt, source lines 10–10. Cross-chapter criterion invoked at proof lines 1122-1125 to convert kappa(s) algebraically closed in kappa(y) into geometric connectedness of the component T; a navigation pointer for a standard criterion, not a replacement of the counted construction.

Definition 0.1. Let S be a scheme. Let X be locally of finite type over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $s \in S$ be a point. A *one step dévissage of $\mathcal{F}/X/S$ over s* is given by morphisms of schemes over S

$$X \xleftarrow{i} Z \xrightarrow{\pi} Y$$

and a quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$ of finite type such that

(1) X, S, Z and Y are affine,
(2) i is a closed immersion of finite presentation,
(3) $\mathcal{F} \cong i_*\mathcal{G}$,
(4) π is finite, and
(5) the structure morphism $Y \to S$ is smooth with geometrically irreducible fibres of dimension $\dim(\mathrm{Supp}(\mathcal{F}_s))$.

In this case we say $(Z, Y, i, \pi, \mathcal{G})$ is a one step dévissage of $\mathcal{F}/X/S$ over s.

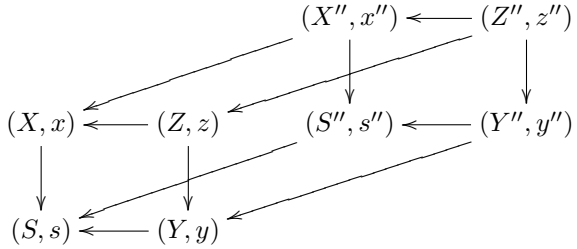
Definition 0.2. Let S be a scheme. Let X be locally of finite type over S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite type. Let $x \in X$ be a point with image s in S. A *one step dévissage of $\mathcal{F}/X/S$ at x* is a system $(Z, Y, i, \pi, \mathcal{G}, z, y)$, where $(Z, Y, i, \pi, \mathcal{G})$ is a one step dévissage of $\mathcal{F}/X/S$ over s and

(1) $\dim_x(\mathrm{Supp}(\mathcal{F}_s)) = \dim(\mathrm{Supp}(\mathcal{F}_s))$,
(2) $z \in Z$ is a point with $i(z) = x$ and $\pi(z) = y$,
(3) we have $\pi^{-1}(\{y\}) = \{z\}$,
(4) the extension $\kappa(y)/\kappa(s)$ is purely transcendental.

Lemma 0.3. *Let S, X, $\mathcal{F}$, x, s be as in Definition 0.2. Let $(Z, Y, i, \pi, \mathcal{G}, z, y)$ be a one step dévissage of $\mathcal{F}/X/S$ at x. Let*

$$\begin{array}{ccc} (Y, y) & \longleftarrow & (Y', y') \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (S', s') \end{array}$$

be a commutative diagram of pointed schemes such that the horizontal arrows are elementary étale neighbourhoods. Then there exists a commutative diagram



of pointed schemes with the following properties:

(1) $(S'', s'') \to (S', s')$ *is an elementary étale neighbourhood and the morphism* $S'' \to S$ *is the composition* $S'' \to S' \to S$,
(2) Y'' *is an open subscheme of* $Y' \times_{S'} S''$,
(3) $Z'' = Z \times_Y Y''$,
(4) $(X'', x'') \to (X, x)$ *is an elementary étale neighbourhood, and*
(5) $(Z'', Y'', i'', \pi'', \mathcal{G}'', z'', y'')$ *is a one step dévissage at* x'' *of the sheaf* $\mathcal{F}''$.

Here $\mathcal{F}''$ *(resp.* $\mathcal{G}''$*) is the pullback of* $\mathcal{F}$ *(resp.* $\mathcal{G}$*) via the morphism* $X'' \to X$ *(resp.* $Z'' \to Z$*) and* $i'' : Z'' \to X''$ *and* $\pi'' : Z'' \to Y''$ *are as in the diagram.*

Proof. Let $(S'', s'') \to (S', s')$ be any elementary étale neighbourhood with S'' affine. Let $Y'' \subset Y' \times_{S'} S''$ be any affine open neighbourhood containing the point $y'' = (y', s'')$. Then we obtain an affine (Z'', z'') by (3). Moreover $Z_{S''} \to X_{S''}$ is a closed immersion and $Z'' \to Z_{S''}$ is an étale morphism. Hence Lemma 057R applies and we can find an étale morphism $X'' \to X_{S'}$ of affines such that $Z'' \cong X'' \times_{X_{S'}} Z_{S'}$. Denote $i'' : Z'' \to X''$ the corresponding closed immersion. Setting $x'' = i''(z'')$ we obtain a commutative diagram as in the lemma. Properties (1), (2), (3), and (4) hold by construction. Thus it suffices to show that (5) holds for a suitable choice of $(S'', s'') \to (S', s')$ and Y''.

We first list those properties which hold for any choice of $(S'', s'') \to (S', s')$ and Y'' as in the first paragraph. As we have $Z'' = X'' \times_X Z$ by construction we see that $i''_*\mathcal{G}'' = \mathcal{F}''$ (with notation as in the statement of the lemma), see Cohomology of Schemes, Lemma 02KG. Set $n = \dim(\mathrm{Supp}(\mathcal{F}_s)) = \dim_x(\mathrm{Supp}(\mathcal{F}_s))$. The morphism $Y'' \to S''$ is smooth of relative dimension n (because $Y' \to S'$ is smooth of relative dimension n as the composition $Y' \to Y_{S'} \to S'$ of an étale and smooth morphism of relative dimension n and because base change preserves smooth morphisms of relative dimension n). We have $\kappa(y'') = \kappa(y)$ and $\kappa(s) = \kappa(s'')$ hence $\kappa(y'')$ is a purely transcendental extension of $\kappa(s'')$. The morphism of fibres $X''_{s''} \to X_s$ is an étale morphism of affine schemes over $\kappa(s) = \kappa(s'')$ mapping the point x'' to the point x and pulling back $\mathcal{F}_s$ to $\mathcal{F}''_{s''}$. Hence

$$\dim(\mathrm{Supp}(\mathcal{F}''_{s''})) = \dim(\mathrm{Supp}(\mathcal{F}_s)) = n = \dim_x(\mathrm{Supp}(\mathcal{F}_s)) = \dim_{x''}(\mathrm{Supp}(\mathcal{F}''_{s''}))$$

because dimension is invariant under étale localization, see Descent, Lemma 04N4. As $\pi'' : Z'' \to Y''$ is the base change of π we see that π'' is finite and as $\kappa(y) = \kappa(y'')$ we see that $\pi^{-1}(\{y''\}) = \{z''\}$.

At this point we have verified all the conditions of Definition 0.1 except we have not verified that $Y'' \to S''$ has geometrically irreducible fibres. Of course in general this is not going to be true, and it is at this point that we will use that $\kappa(s) \subset \kappa(y)$ is purely transcendental. Namely, let $T \subset Y'_{s'}$ be the irreducible component of $Y'_{s'}$ containing $y' = (y, s')$. Note that T is an open subscheme of $Y'_{s'}$ as this is a smooth scheme over $\kappa(s')$. By Varieties, Lemma 04KV we see that T is geometrically connected because $\kappa(s') = \kappa(s)$ is algebraically closed in $\kappa(y') = \kappa(y)$. As T is smooth

we see that T is geometrically irreducible. Hence More on Morphisms, Lemma 057J applies and we can find an elementary étale morphism $(S'', s'') \to (S', s')$ and an affine open $Y'' \subset Y'_{S''}$ such that all fibres of $Y'' \to S''$ are geometrically irreducible and such that $T = Y''_{s''}$. After shrinking (first Y'' and then S'') we may assume that both Y'' and S'' are affine. This finishes the proof of the lemma. $\square$